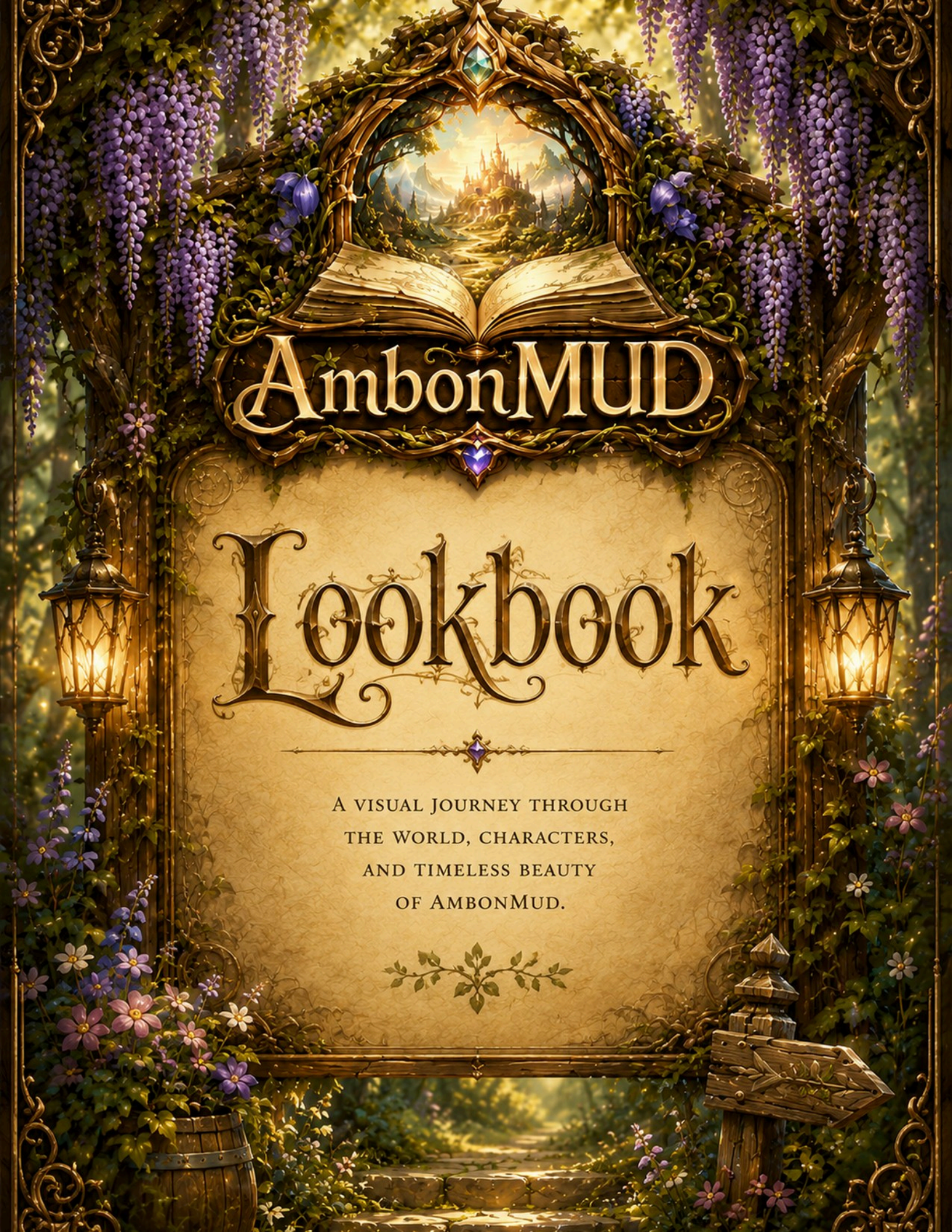

AmbonMUD
Lookbook
A visual Journey through
the World, characters,
and timeless beauty
of AmbonMud.

AmbonMUD Lookbook

A visual tour of AmbonMUD — a cozy adventure game set in the world of Ambon, where players are called from beyond the veil to become its heroes.

The Game at a Glance

AmbonMUD is a multiplayer fantasy world in the classic MUD tradition, rebuilt for the browser: every room, character, and panel is rendered over painted scenes, while the same world remains fully playable over plain telnet. New players can jump in instantly as a guest with the demo mode and claim their character later, or create an account in a painted login flow that walks them through choosing one of **nine races** and **six classes** (Bulwark, Arcanist, Veil, Herald, Scrollwright, and Songweaver).

Adventure & progression. Characters level by exploring and fighting through hand-authored zones, from the 128-room Auringold Academy tutorial onward. Class trainers teach new abilities with earned skill points, and multi-classing lets a character branch into additional classes over time. Abilities, status effects, and tactical choices (including a wimpy auto-flee threshold) drive real-time, tick-based combat with damage toasts, enemy vitals, and a combat log. Beyond fighting, a complete **pacifist path** (the Akathavae) progresses through illumination and a personal Arcanum journal instead of violence. Rare cosmetic mob variants — albino, ember, shadow-touched, spectral, and kin — appear on a small chance for collectors and bragging rights.

Quests & goals. NPCs offer branching, voice-acted dialogue with structured quest offers; the quest log tracks active, daily, weekly, bounty, and global quests. Achievements, leaderboards, factions with reputation tiers, and special currencies (quest points, honor, crafting tokens) give long-term goals past the level cap.

Economy & crafting. Shops buy and sell with configurable economy multipliers; the bank stores gold and items; an auction house runs player-to-player sales; direct trading handles face-to-face deals. Gathering nodes feed a recipe-driven crafting system, with enchanting to improve gear and item manuals to study what you've found.

Places & possessions. Player housing is bought through realtors and furnished; guilds get shared halls. Inns offer rest, pets can be adopted and named, the stylist re-styles your look, and a wardrobe manages outfits.

Diversions & world life. Lottery drawings, dice tables, jukeboxes with zone soundtracks, puzzles and riddle doors, instanced dungeons, and PvP dueling round out the world — which itself lives on a day/night clock with seasons, weather, and scheduled world events that change the sky overhead.

Together. Groups for shared adventuring and XP, guilds with boards and halls, friends lists, mail with attachments, global and group chat channels, and a who's-online board keep it social.

Aineron
Lustriae
Estarion
Aetherion
Aetherae
Lustraon
Lithae
Eqreon
Faegrimm
Kristolith
Kaerinlith
Thalnakh
Caldera
Ophirae
Tessikon
Calderon
Nakhara
Thalvorek
Pyrae
Kiriton
Mycorae
Archae
Mycollum
Fenspore
Veratharon
Yaleron
Vulgorth
Kitsarae
Saeloria
Marldon
Lyrasael
Voxspore
Kitsalora
Sylflorae
Cindaraon
Harroven
Sylvarden
Graktuk
Malukae
Florensyl
Doktuk
Aureliae
Lagauri
Radimaren
Phosmaren

Under the Hood

Server. Kotlin on JDK 21. A single-threaded game engine advances the world on a 100 ms tick; all gameplay state lives inside the engine, and the engine speaks to the outside world only through typed inbound/outbound event buses — transports contain no game logic, and the engine contains no transport code. The same binary runs as a single **standalone** process or splits into **gateway** (transports) and **engine** (simulation) processes connected over gRPC, with Redis-backed buses, optional player caching, and zone sharding/instancing hooks for horizontal scale.

Protocols. Plain telnet for classic clients, and WebSocket with **GMCP** packages for the web client — room state, vitals, maps, shop/trainer/bank inventories, dialogue, quests, and painted-art asset URLs all stream as structured GMCP messages emitted by the server.

Web client. React 19 + PixiJS. The world view is a canvas scene seated over painted room art; UI panels are DOM, skinned by a *painted panel contract*: the server registers art asset keys, delivers URLs via GMCP, and the client locks each panel's aspect ratio to the art and seats live controls onto the painting with percentage insets — with a CSS-only fallback when art is absent.

Gameplay systems. A sealed `Command` hierarchy of roughly two hundred verbs parses into typed inbound events that a thin router dispatches across three dozen focused handlers — combat, quests, crafting and enchanting, housing and guilds, trading and the auction house, pets, dueling, and the rest. NPCs run behavior trees and branching, voice-acted dialogue; class trainers, multi-classing, abilities and status effects, factions with reputation tiers, achievements, and special currencies all layer onto the same tick loop. A complete pacifist progression path advances through illumination rather than combat, sharing the engine with everything else.

Realtime delivery. Outbound text and GMCP updates pass through a router that applies backpressure and coalesces consecutive prompts before they reach a session, with a final-message-then-close path for clean disconnects. The same engine code runs unchanged whether the event bus behind it is in-process (standalone), Redis-backed (multi-process), or gRPC (gateway↔engine) — the engine never learns which.

World as data. Zones, rooms, mobs, items, shops, trainers, quests, recipes, puzzles, and dungeons are declarative YAML validated by a strict loader. The world ships separately from the binary: production assembles its dataset (config overlay + zone files + sprites + achievements) from object storage at boot, so content updates deploy without rebuilding the server. The companion desktop app **Arcanum** (Tauri + React) edits the same schema — entity editors, validation mirroring the server's rules, AI art and ElevenLabs voice generation with content-addressed publishing to R2/CDN.

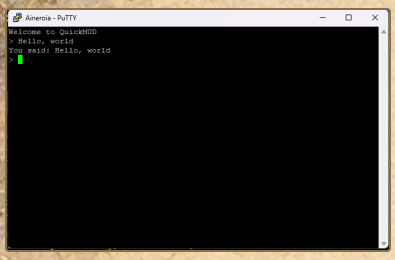
Persistence. Pluggable player store — YAML files with atomic writes for simple deployments or PostgreSQL (Flyway-migrated) for production — wrapped in a write-coalescing worker and an optional Redis cache. Schema changes ship as Flyway migrations; a coverage test fails the build if a persisted field is added without its mapping on both sides.

Testing. The suite runs on a deterministic injected clock and in-memory repositories, so time and persistence are reproducible; database tests use H2 in PostgreSQL mode with no Docker required. Coroutine tests advance virtual time rather than sleeping, and the world loader is exercised against fixtures of both valid and deliberately invalid zone files to keep validation strict.

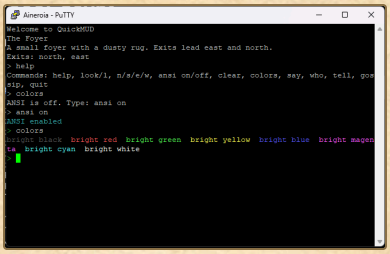
Operations. Docker images deployed via GitHub Actions with health-checked restarts, an admin HTTP API for live reloads and moderation, Prometheus metrics, and CI running lint, unit, and integration suites on every PR.

From Telnet to Painted

AmbonMUD began as a plain telnet proof-of-concept and grew, across six generations of client, into the fully-painted interface this book tours. The whole journey, oldest to newest:



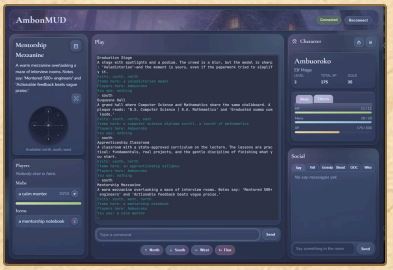
v0 — Telnet (PuTTY): the first proof of concept



v0.5 — Telnet with ANSI color

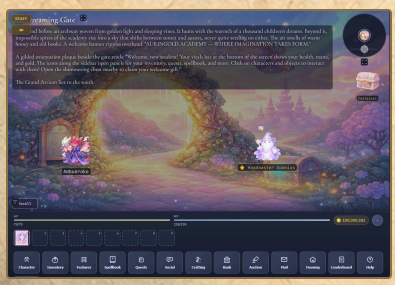
v1 — First web terminal, no panels

v2 — Web client with panels



v3 — Surreal Gentle Magic redesign

v4 — PixiJS canvas and JRPG scenes



v5 — Painted scenes, glass-UI panels

v6 — Fully painted panels (current)

The Six Classes of Ambon

Bulwark

Immovable defender clad in heavy armor, holding the line against overwhelming odds.

Arcanist

Master of advanced spellcraft who bends illusion and reality through communion with the Arcanum.

Veil

Stealth assassins who strike from shadow and vanish before their enemies can react.

Herald

Divine conduit who channels patron power to heal wounds, restore strength, and kindle hope amid darkness.

Scrollwright

Craftspeople who command enchanted clockwork constructs powered by captured starlight and celestial magic.

Songweaver

Minstrels and bards who channel primal magic through melody, narrative, and performance, shaping reality itself.

The Peoples of Ambon

The Children of the Creator

Woven into being by the Creator at the dawn of creation.

Archae

The first children of the Creator, adaptable and ambitious inheritors of divine legacy.

Shaped by Kvarcgagap

The primordial god of bedrock, gems, and the glitter beneath the ground.

Lithae

Living beings of crystal and stone, their forms shimmering with the earth's raw beauty.

Made by the Naraxian Akathavae

Raised from animals who sought cultivation and enlightenment.

Kitsarae

Animal spirits who walk between animal and humanoid form, masters of deception and subtle enchantment.

The Children of Aineroia

The goddess who rose from a mortal Archae healer.

Lustriae

Eternally youthful fae who perceive time as a living current and flit between moments with curious playfulness.

Aetherae

Beings of shadow and mist born from Aineroia's own shadow — shapeshifters of the far north who tend the Luneqrae.

Pyrae

Children of the Phoenix, beings of living flame born from stories of rebirth and resurrection.

Ophirae

Serpentine abyssal dragons who rule the cold northern waters from underwater cities built of leviathan bone.

Mycorae

Fungal people grown from living mycelium, embodying the natural cycle of decay and rebirth.

Aureliae

Bioluminescent beings of the deep ocean, drifting like living constellations through Ambon's darkest waters.

Creatures of Ambon

Malukae

Ambon's trolls, fashioned from the cooling malice of Malkuth's heart — stone-skinned and cold-blooded, walled onto a sun-baked peninsula and slowly dwindling. The one people no adventurer can play.

Sylflorae

A people accidentally sown when the fading Alorae spent their light growing crops. Driven underground in the Kirot War, they thrive in cavern gardens fed by Astriel, their harvest-god, and lit by the diminished remnant of Kvarcgagap — the old gem-god worn down to a gleam in the crystal.

Animae

Fully clockwork folk wrought by Tessikar the Springwright — brass-and-light automatons of turning gears and glowing filigree, an itinerant tinker culture out of clockwork Tessikon.

Luneqrae

Moonlight horses descended from Aineroia's youngest child, bound to run on moonlight across the northern plains in the care of the Aetherae.

Alorae

The luminous Divine Heralds who descended with the Creator. Their angelic power all but spent, they endure as a scattered remnant, giving the last of their light to any who need it.

Figures of Ambon

Aineroia

An Archae herbalist taken as bride by the tyrant Malkuth; when she found him draining the life of their children, she slew him with the obsidian dagger and ascended to godhood. The mother-goddess of many of Ambon's peoples.

Malkuth

A cruel battle-mage tyrant from beyond the veil who wed Aineroia only to secretly drain the power of their children. Slain by her hand, his death tore a caldera into the world.

Krioshaeu

The kind face Aineroia gave to Death — a gentle woman who meets each soul and leads it onward, turning a formless terror into a comforting guide.

Naraxian Akathavae

Keeper of the divine history and one of the three primordial powers — the Word, in whose native universe Ambon was made. The many-faced god of scholars, musicians, and stories who records all that is into the Arcanum, gave music to the Archae, and cultivated the Kitsarae from seekers among the beasts.

Kvarcgagap

An ancient primordial being from beyond Ambon — god of bedrock, caverns, gems, and ore — who flows through solid stone as shimmering opal quicksilver. Shaper of the Lithae.

Caemyr

A visiting scholar-god who helped to repair the arcane scar left by the ascension of Aineroia. Mentor to Ambuoroko.

Verathaios

The World Healer, a gentle god of growth who arrived from beyond the veil after Aineroia's ascension; his divinity is the quiet miracle of life taking root.

Maelsang

A god of storms and seas from another world, heralded by lightning across cloudless skies and waves that sing with otherworldly voices.

Ambuoroko

Ambuoroko Ambonien, King of Ambon — an Aetherae who mastered illusion across centuries in Caemyr's timeless Athenaeum Unbound and came home a god of law and justice: the beggar at the gate whose judgment is final, who rewards the kind and curses the cruel.

Elustriaine

The Faerie Queen of the Lustriae, her crown woven of crystallized moments and captured dawns, bound to duties that stretch across the fabric of time.

Tessikar

The Springwright, a clockwork artificer born beyond the veil who arrived through a temporal-mechanical gateway of his own making — patron of brass-and-starlight machinery.

Astriel

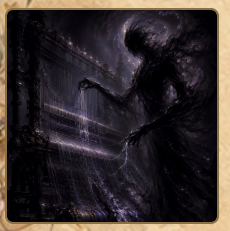
An Alorae Divine Herald turned god of the harvest — his cultivation-magic grew the crops that fed the struggling Archae, and from that same craft the Sylflorae were accidentally sown.

Kirot Aethiryn

An Alorae Divine Herald who turned from the light to dark necromancy — the sole master of death-magic in Ambon, walking a path no other dares.

Sorayangriel

An Alorae who, when the Creator's light withdrew and the world fell cold and dark, burned away all her divine power at once to become the sun — ascending instead of dying. Ambon's goddess of light, fire, and warmth, robed in living flame and dwelling within the sun by day.



Braelaatum

One of the three primordial powers — the silence before creation. A goddess of nothing: pitch dark, freezing cold, the unraveller of looms, who judged the empty void already perfect and would pull all that her siblings made back into it. Sealed in her own realm, she touches Ambon only through cold, shadow, and the hands that would let her in.

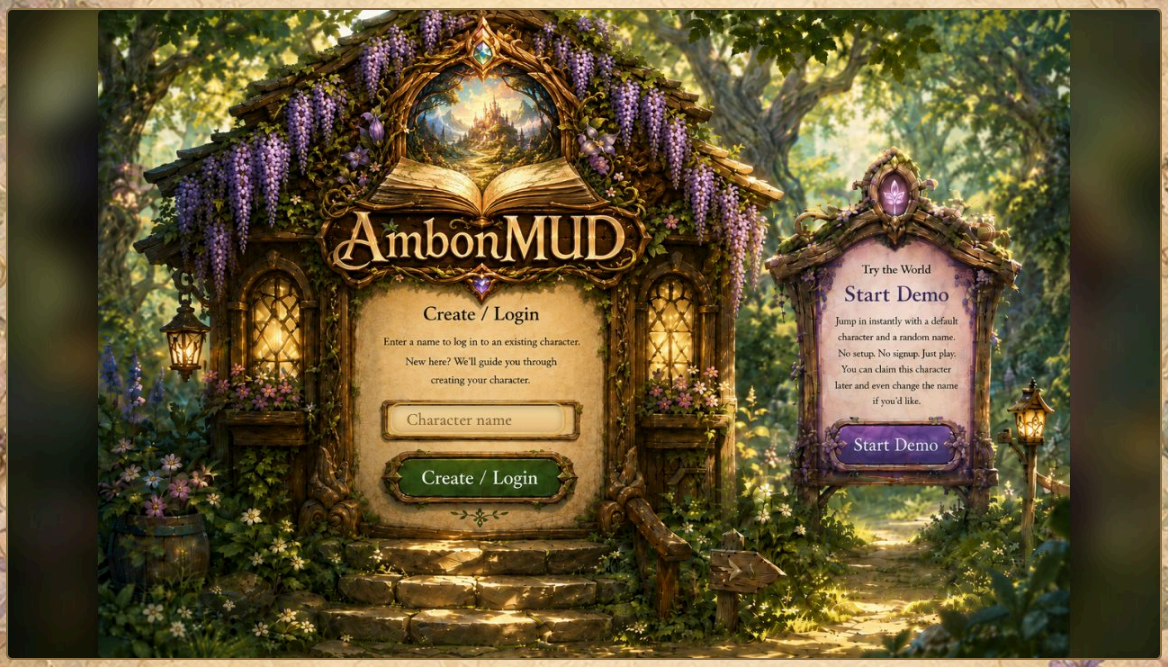
Kolksprut

Trickster, traveler, and gambler — a cuttlefish who cultivated his way from beast to godhood on sheer thirst for knowledge, then won immortality from the Naraxian on a single coin-flip. A shapeshifting weaver of illusions and god of luck.

The Drengariae

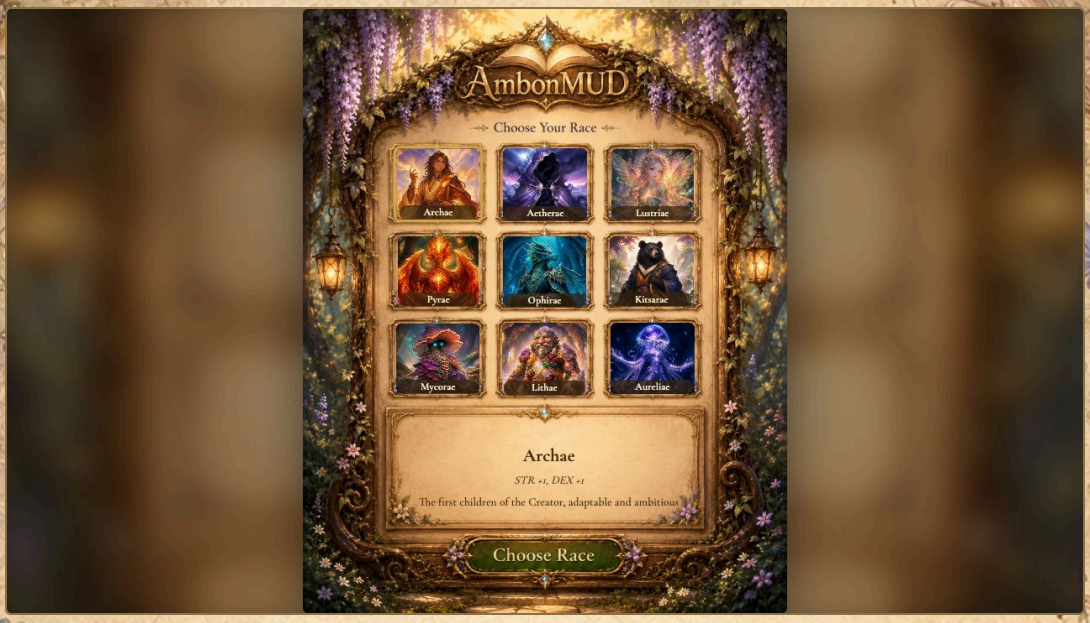
Alorae warriors who ringed the first settlement with flaming swords through the Long Dark, holding back the cold and the creatures until they froze where they stood. Sorayangriel's first dawn melted them away — and they answer the call to war now as a single multi-faced god of frost and fire, splitting into its fallen and merging again.

Name entry



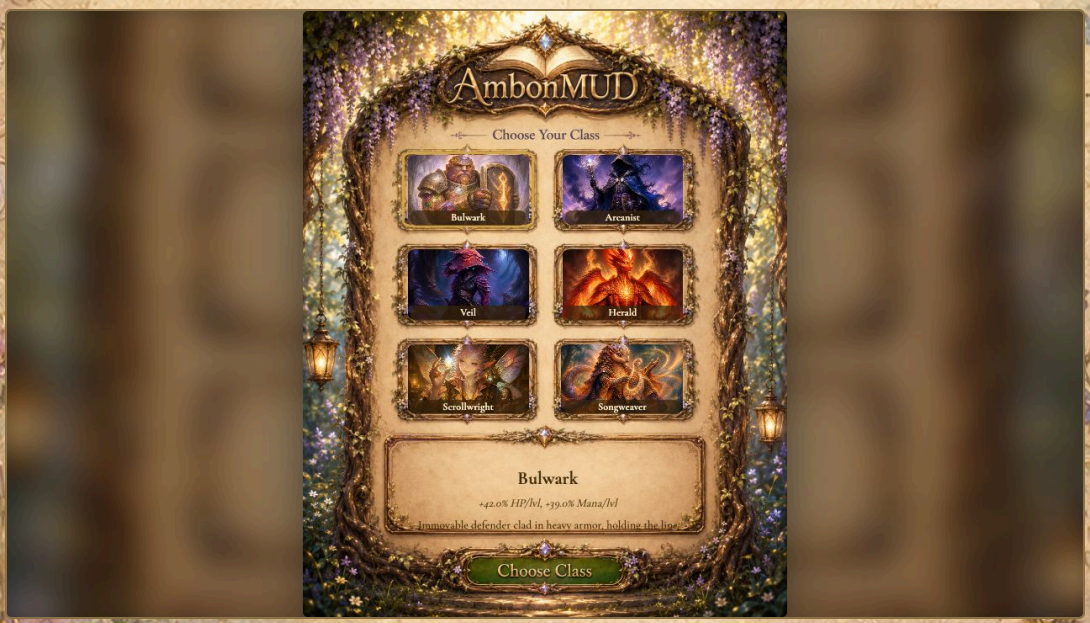
Name entry — the cottage login scene with the Start Demo side panel

Race selection



Race selection — nine races of Ambon in their painted niches

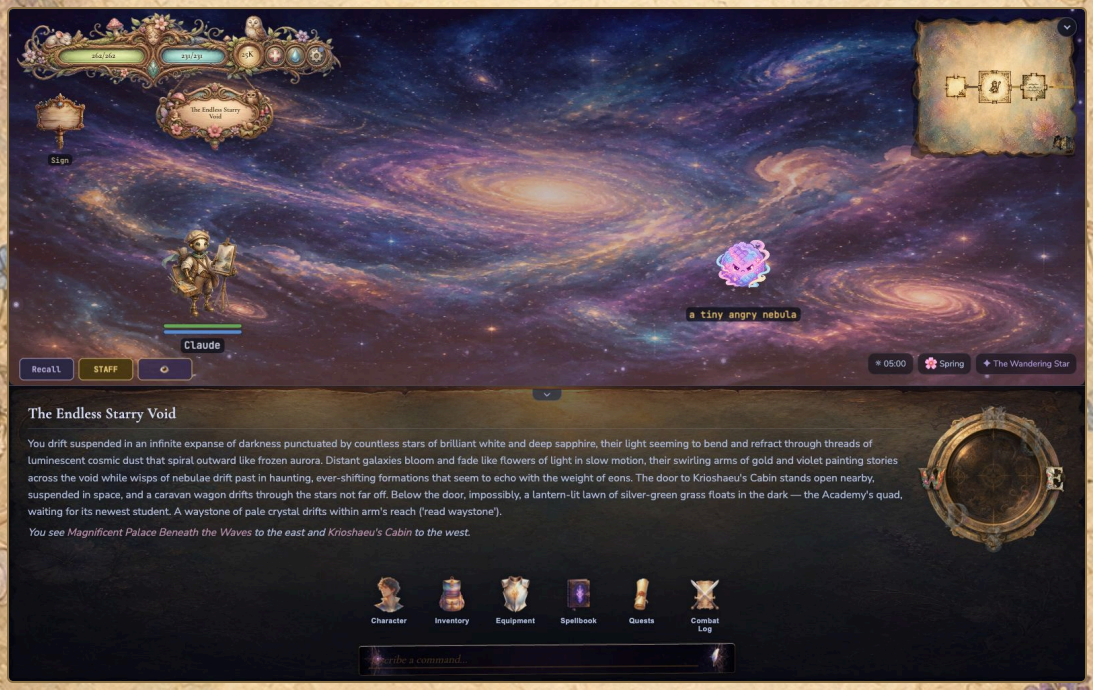
Class selection



Class selection — the six classes

The World View

The Adventurer's View



The in-game view: painted room scene with player and NPC sprites, vitals HUD, minimap, room narrative, exits compass, and the panel dock.

The vitals bar, up close

The vitals bar, carved into the flowering branch that frames the scene: health and mana pools, gold, and one-tap heal and mana potions.

The room sign

Every room hangs its name on a painted placard — tap it to enlarge.

Character & Progression

Character sheet

Character sheet — portrait, stats, settings, and links to professions and achievements

Professions

Professions

Achievements

Achievements

Inventory

Inventory

Equipment

Equipment

Trainer

The class trainer at the Library of Fairytales — spend skill points to learn abilities

Spellbook

Spellbook — learned abilities with hotbar binding

Quest log with an active quest

Quest log with an active quest

Chat board

The Social Board, mid-conversation on the server-wide Gossip channel — players coordinating a catacombs run. Five channels (Say, Tell, Gossip, Shout, OOC) share the one painted frame.

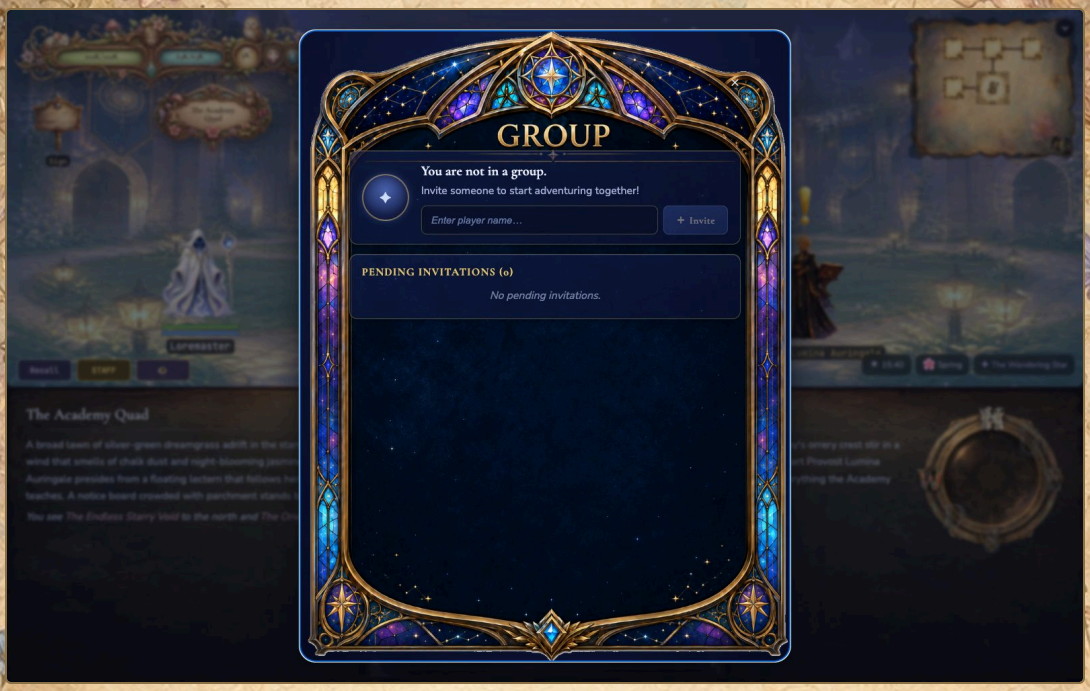
Social

Who's online

Friends

Guild board

Group board



Mail

Help compendium

Shop

Buying and selling at Madam Daydreams' General Store

Commerce & Services

Bank vault

The Bursar's Vault — gold and item storage

Auction house

The Auction Hall of Whispered Bids — player-to-player sales

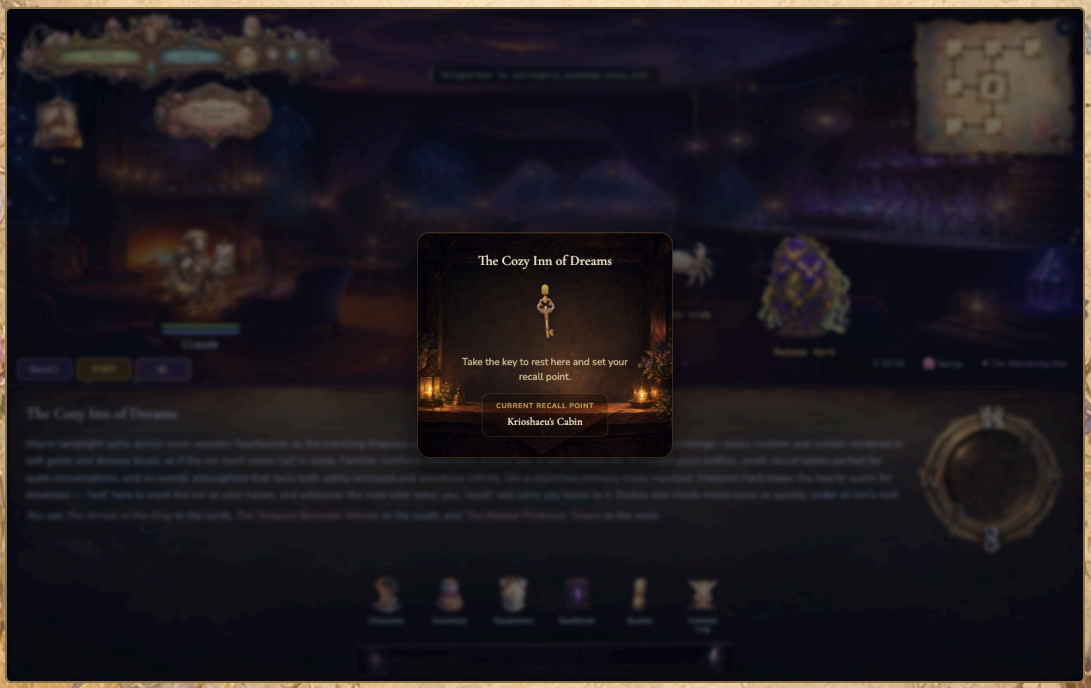
Stylist

The Twilight Salon — cosmetic restyling

Housing realtor

The Estates Broker's Office — buy and furnish a home

Inn & recall



The Cozy Inn of Dreams — take the hanging key to rest and make the inn your recall point, and the `recall` command will always carry you home to it

Crafting recipes

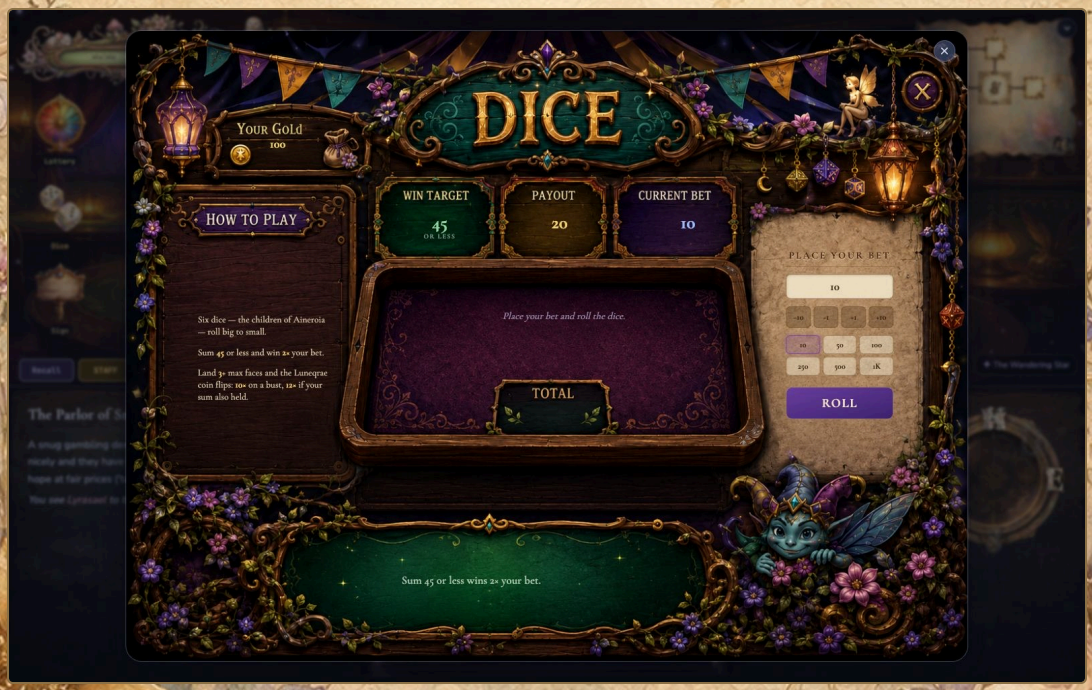
The recipe book at the Crafters' Workshop

Games & Diversions

Lottery

The lottery board in the academy's games parlor — jackpot, ticket numbers, and the nightly drawing

Dice table



Kolksprut's honest dice in the Parlor of Small Fortunes

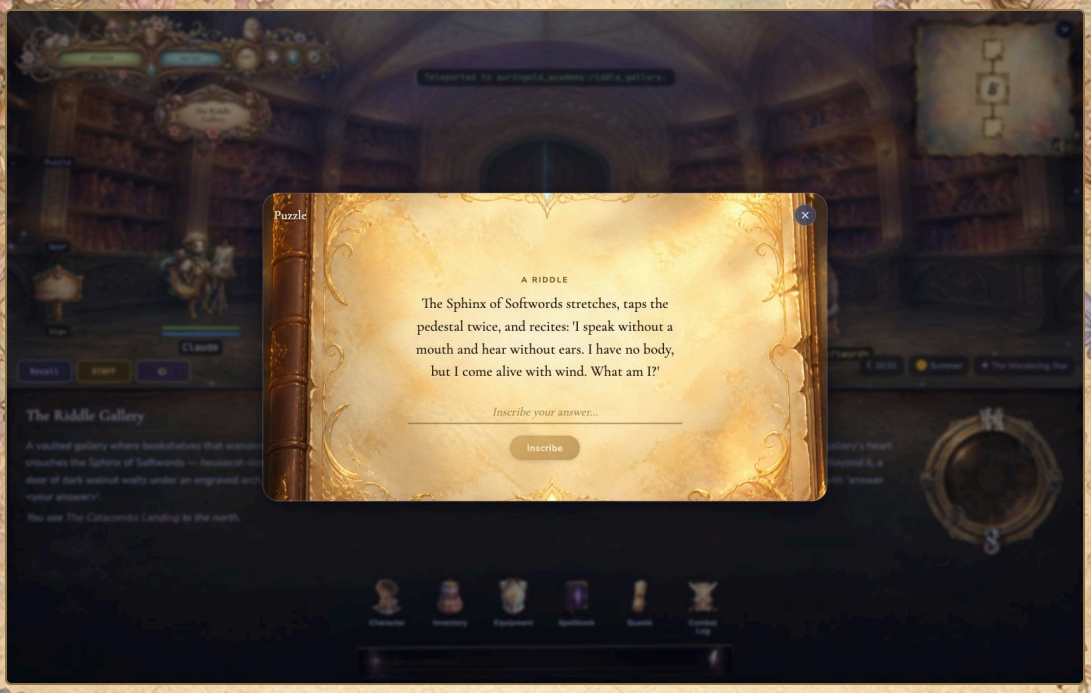
Jukebox

The jukebox in the Parlor of Borrowed Songs — pay a few coins and your chosen track plays for everyone in the room

Music box

The music box in the Fortune Teller's Astral Parlor — a free, one-song miniature of the jukebox. Wind it up and "The Word of Ambon" plays only for you, its lyrics scrolling past as the song follows you from room to room. Plus you keep a souvenir: the first play mints a collectible lyric sheet into your inventory — a little printed keepsake of the song you can read back any time

Puzzle



A riddle from the Sphinx of Softwords in the Riddle Gallery — inscribe the answer to open the way

Treasure chest

A painted container in Kvarcgagap's Vault — open the chest and take the loot inside

NPC Dialogue, Quests & Combat

NPC dialogue

Branching conversation with Krioshaeu, who offers the first quest

Combat

Facing the storm-god A Dream of Maelsang in the Tempest Between Worlds — damage toasts, enemy vitals, and the flee escape hatch

Combat continues

Abilities on the hotbar fire as the bout plays out

Staff Tools

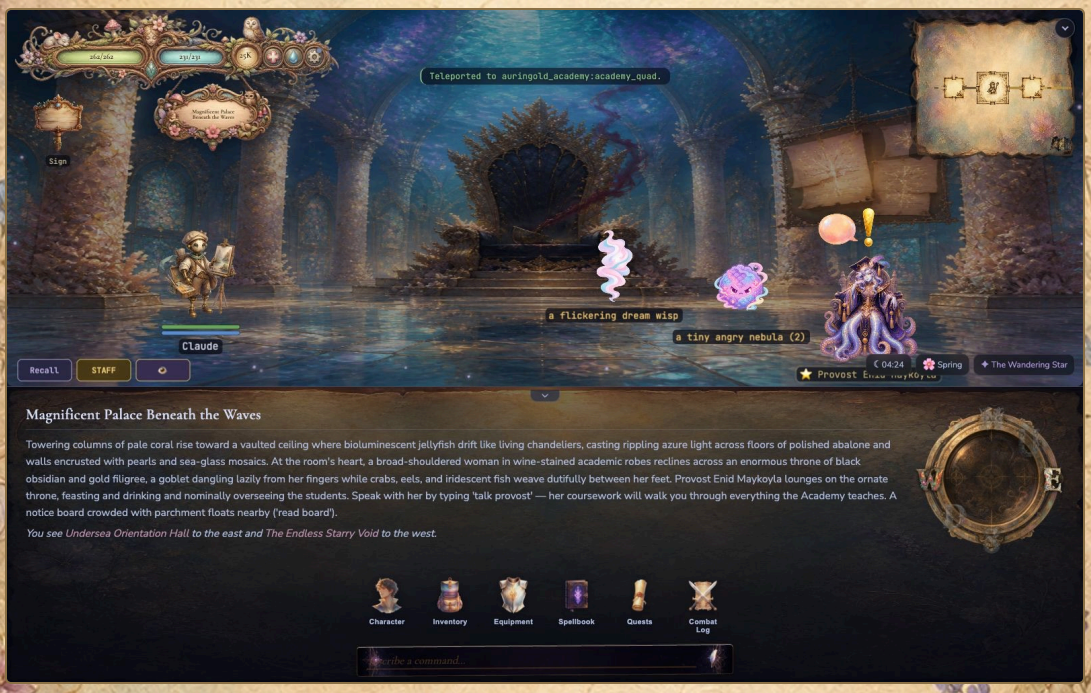
Staff control panel

The staff control panel — teleport, spawn, set level and stats, moderate players, and reload the world live, all gated to staff accounts

A Few Academy Rooms Up Close

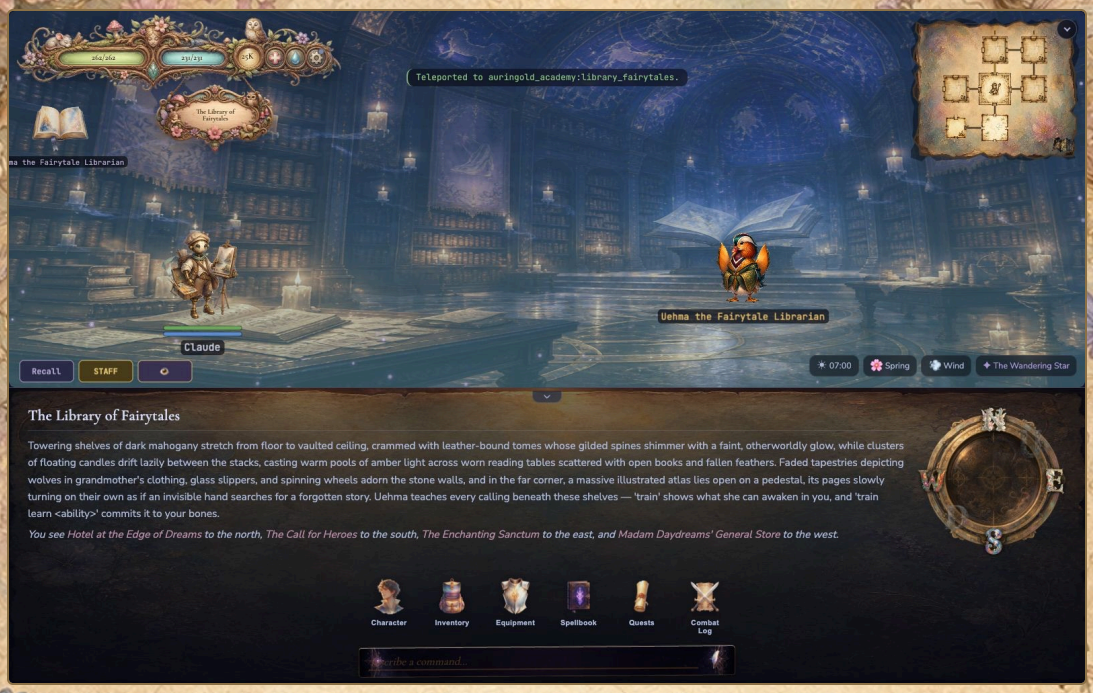
A handful of Auringold Academy rooms at full size — including the homes of services the web client surfaces through the world itself.

Magnificent Palace Beneath the Waves



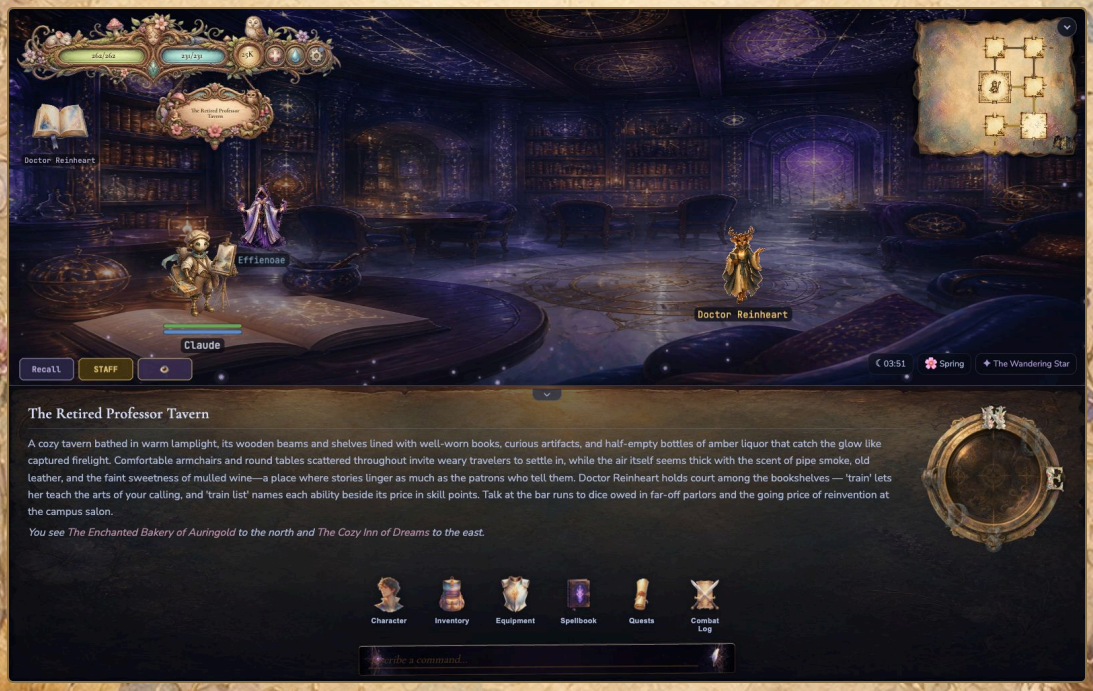
The Academy's heart, reimagined beneath the waves — Provost Enid Maykoyla holds court from a throne of obsidian and gold beneath jellyfish chandeliers

The Library of Fairytales



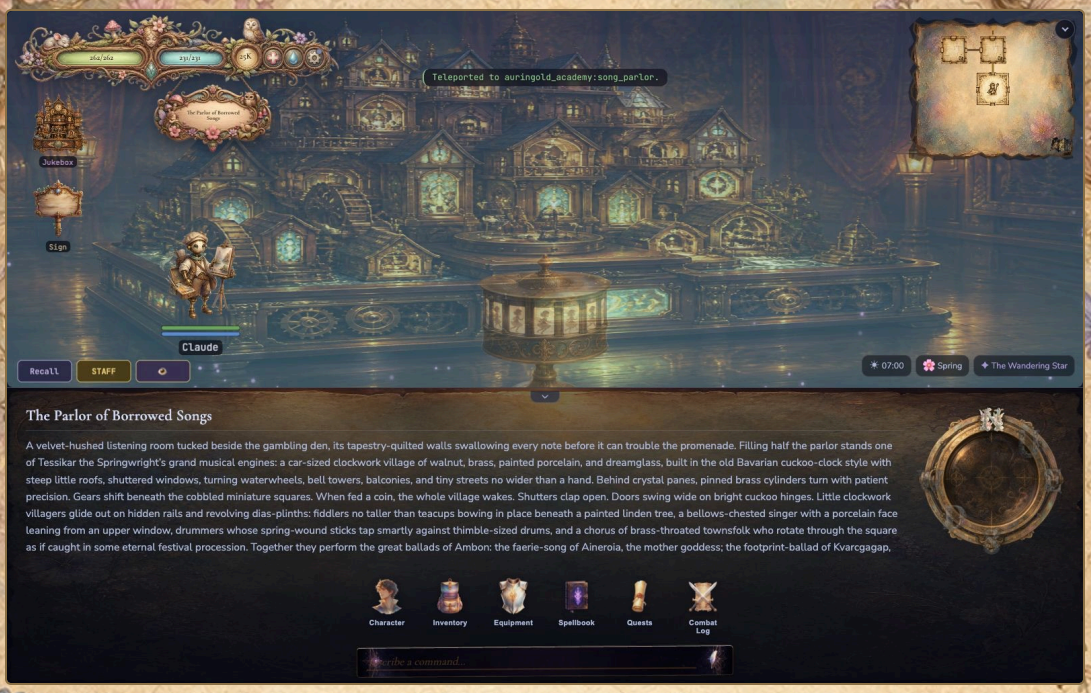
Where the class trainer teaches new abilities for skill points

The Retired Professor Tavern



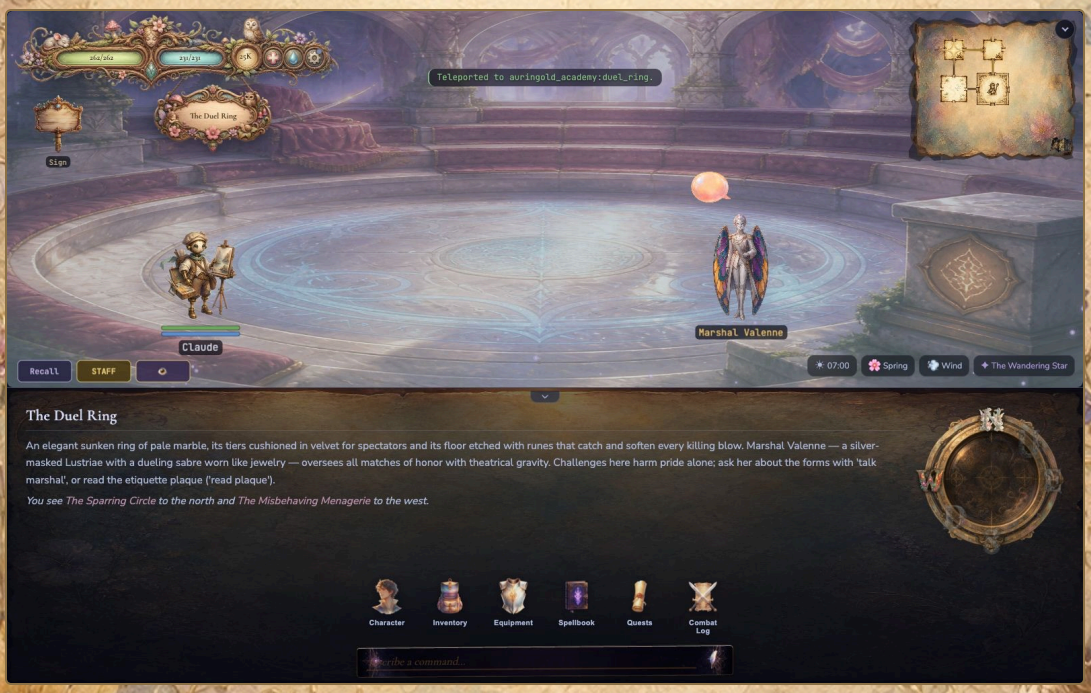
Rest and recover at the Academy's inn

The Parlor of Borrowed Songs



Home of the jukebox that plays for the whole room

The Duel Ring



Where students settle matters with sanctioned PvP duels

The Gathering Garden

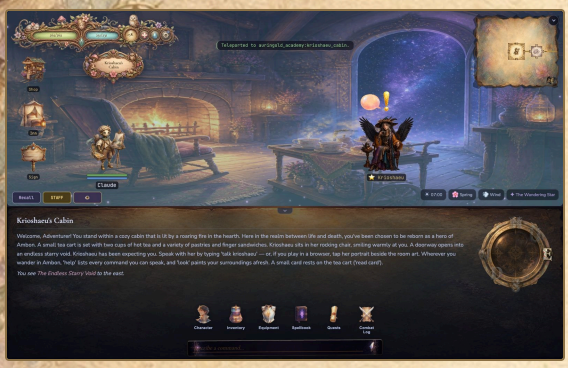
Gathering nodes feed the crafting professions

The Almanac Terrace

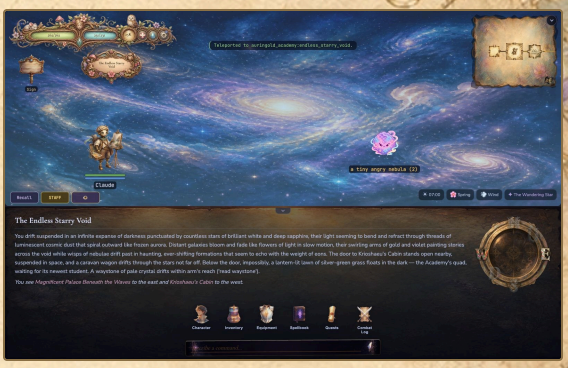
Where the Academy's day/night clock, seasons, and weather turn overhead

The Auringold Academy — All 128 Rooms

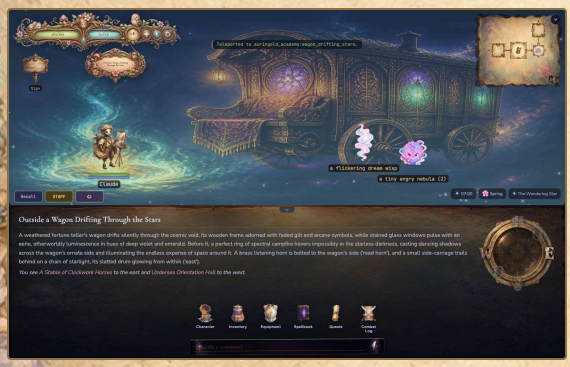
Every room of the Auringold Academy tutorial zone, in zone-file order.



Krioshaeu's Cabin



The Endless Starry Void



Outside a Wagon Drifting Through the Stars

The Zoetrope Carriage

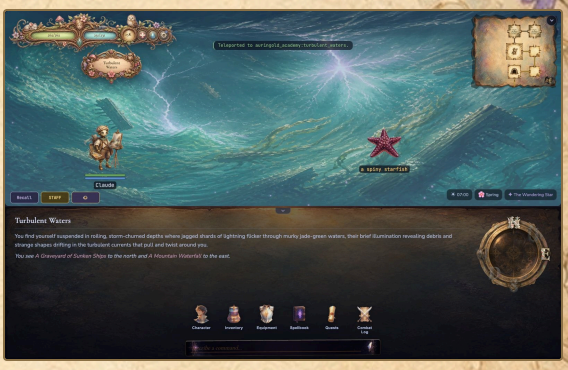
A Memory of Creation

Fortune Teller's Astral Parlor

The Heralds of Creation

The First Archae

The Tempest Between Worlds



Turbulent Waters

A Sandy Shoal

The Foothills Below Kaerinlith

A Sylflorae Farm

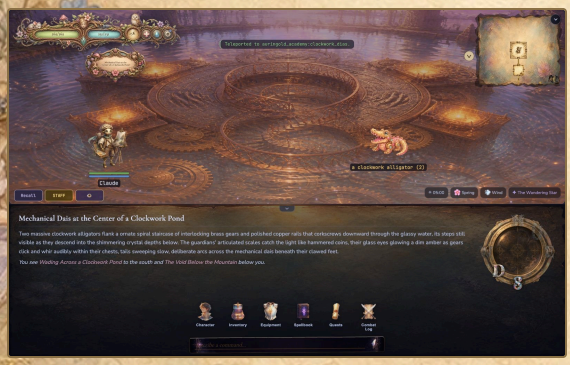
A Clockwork Forest

Deeper in the Clockwork Forest

A Clearing in the Clockwork Forest

The Springwright's Workshop

Beside the Clockwork Pond



Mechanical Dais at the Center of a Clockwork Pond

The Opening of a Shining Crystal Mine

A Shining Crystal Mineshaft

The Void Below the Mountain

Cavern of Sleeping Stone Lithae

The Arrival of the King

The World Seed

The Great Betrayal

The Clockwork Battle

The Birth of the Obsidian Dagger

The Fortune Teller's Prophecy

The Emberfell

A Humble Forest Cabin

The Royal Wedding

In a Tiny, Brightly-Lit Graveyard

The Caldera

Inside the Arcane Tempest

Verathaios's Workshop

Aolexia

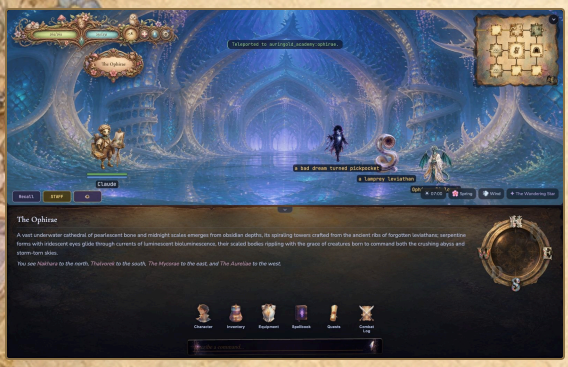
Aolia

The Shattered Fae Wood

The Lustriae

The Pyrae

The Aureliae



The Ophirae

The Mycorae

A Shrieking Night Wind

The Luneqrae

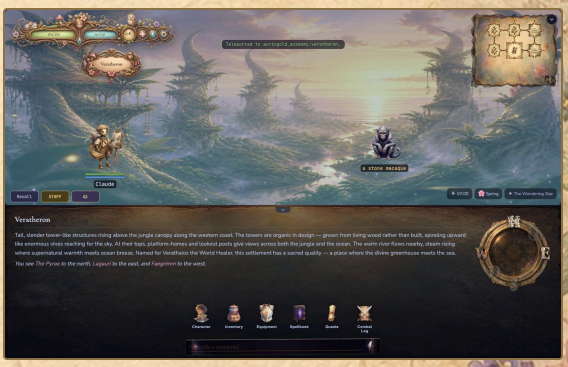
Aineroia Lights the Night Sky

The Aetherae

The Call for Heroes

The Hot Desert Sands

Aineron



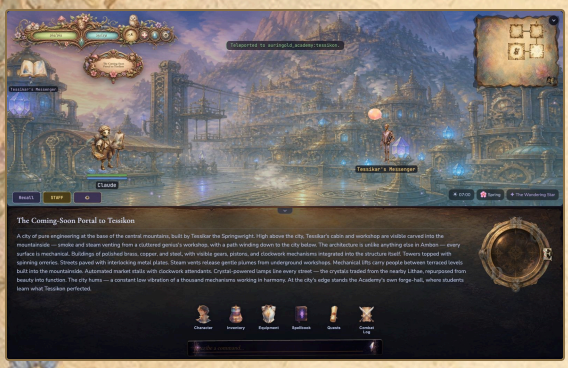
Veratheron

Phosmaren

Thalvorek

Mycollum

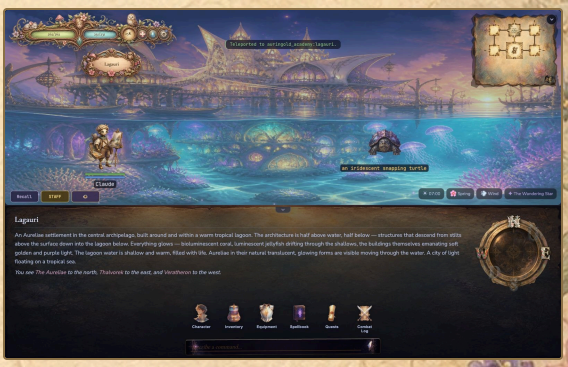
The Creeping Darkness



The Coming-Soon Portal to Tessikon

Faegrimm

Cindaron



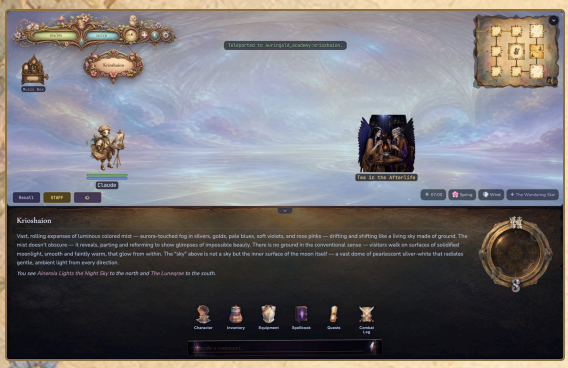
Lagauri

Nakhara

Voxspore

Lyrasael

Vulgorth

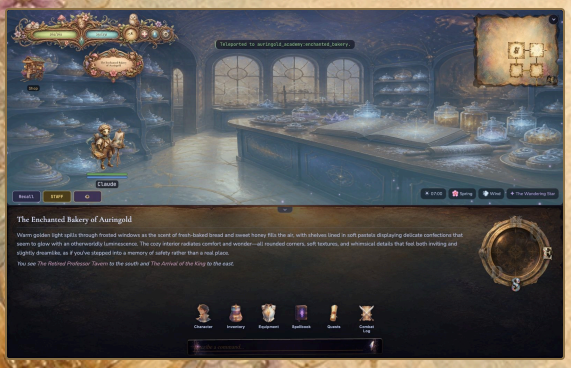


Krioshaion

The Path to Kaerinlith

The Cozy Inn of Dreams

A Mineshaft Full of Gemstones

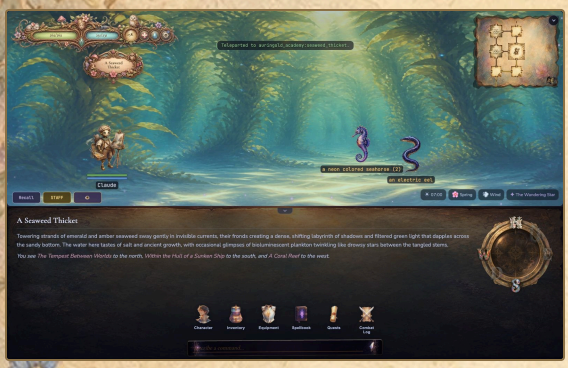


The Enchanted Bakery of Auringold

The Retired Professor Tavern

A Graveyard of Sunken Ships

A Coral Reef



A Seaweed Thicket

Within the Hull of a Sunken Ship

Kvarcgagap's Vault

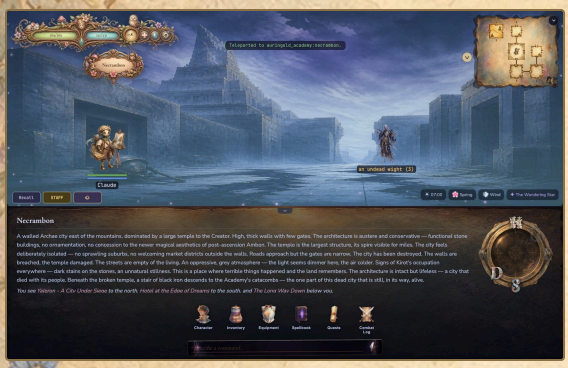
Wading Across a Clockwork Pond

A Stable of Clockwork Horses

The Path to Enlightenment

Aineroia's Cottage

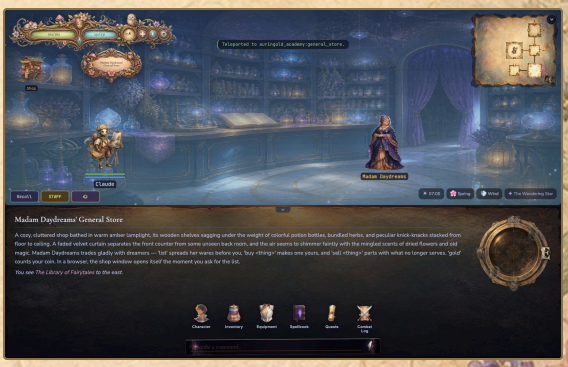
Yaleron - A City Under Siege



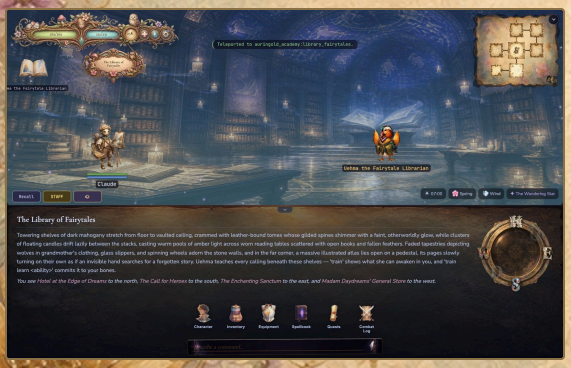
Necrambon

A Resistance Camp in the Mountains

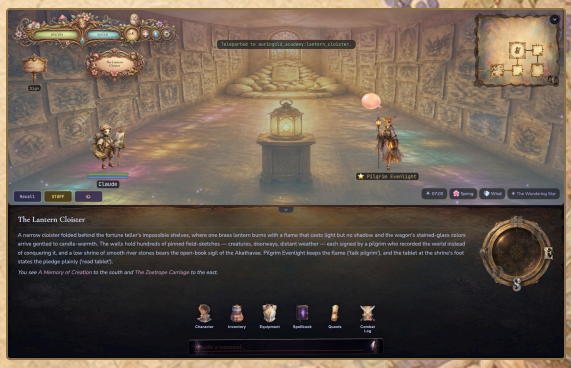
Hotel at the Edge of Dreams



Madam Daydreams' General Store



The Library of Fairytales



The Lantern Cloister

Harsh Kiriton Winter

An Underground River

A Mountain Waterfall

Crafter's Workshop

The Betrayer's Study

The Awakening of the Kitsarae

Magnificent Palace Beneath the Waves

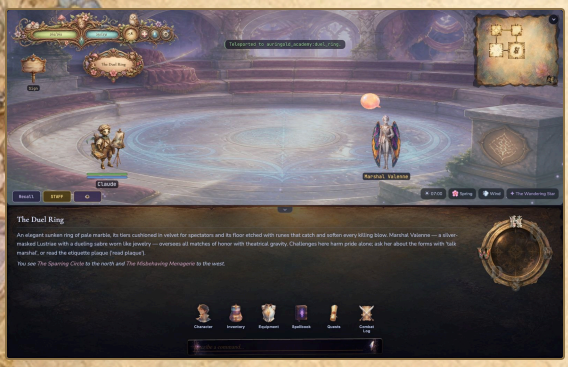
Undersea Orientation Hall

The Proving Yard

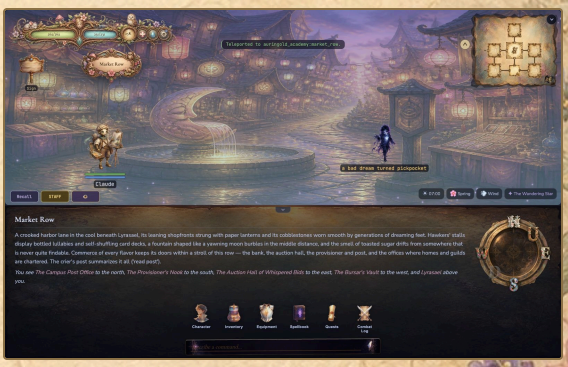
The Sparring Circle

The Misbehaving Menagerie

The Spotters' Walk



The Duel Ring



Market Row

The Bursar's Vault

The Auction Hall of Whispered Bids

The Provisioner's Nook

The Campus Post Office

The Estates Broker's Office

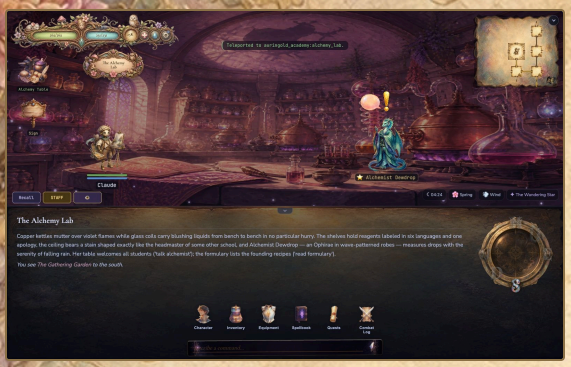
The Guild Registry

The Gathering Garden

The Almanac Terrace

The Old Mine Gate

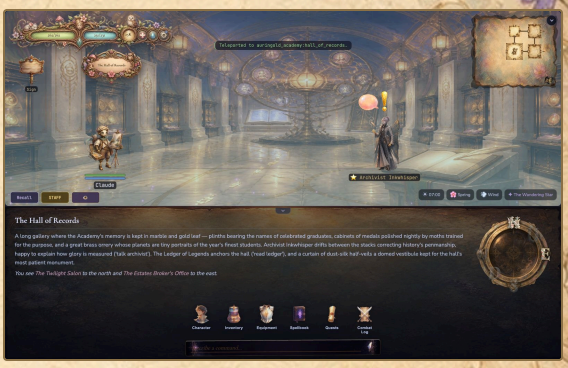
The Crystal Seam



The Alchemy Lab

The Tinker's Bench

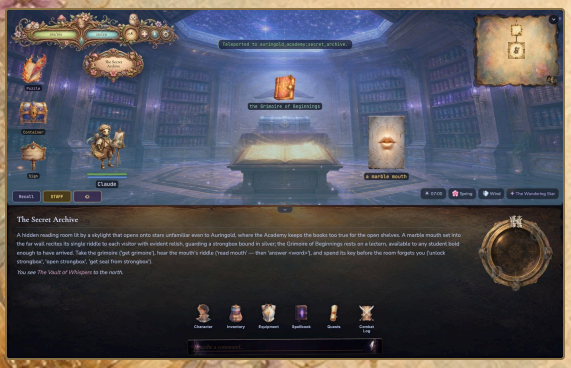
The Enchanting Sanctum



The Hall of Records

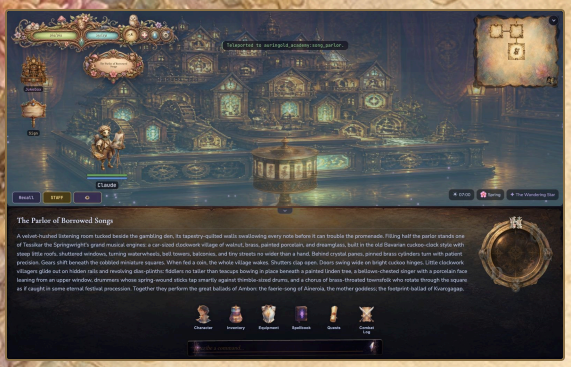
The Riddle Gallery

The Vault of Whispers



The Secret Archive

The Parlor of Small Fortunes



The Parlor of Borrowed Songs

The Twilight Salon

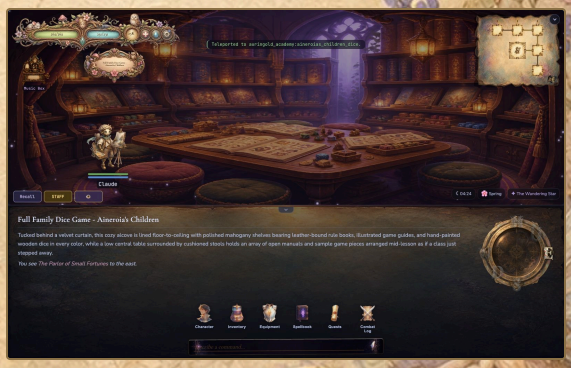
The Long Way Down

The Long Dark Hall

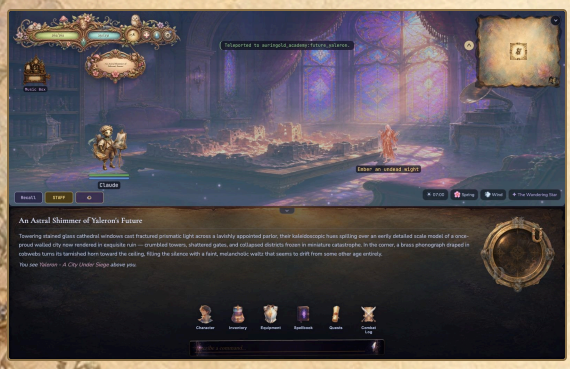
The Chapel of the Last Lamp

The Memento Mori Shrine

The Waiting Dark



Full Family Dice Game — Aineroia's Children



An Astral Shimmer of Yaleron's Future

A Nightmare in the King's Chamber

AmbonMUD